

Coding & Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	Voice-Based concept Understanding Analyser
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/your-team/Voice-Basedconceptunderstandinganalyser
Programming Language(s)	Python
Framework(s) Used)	Streamlit, OpenAI Whisper, Sentence-Transformers (SBERT), Librosa, SQLite, ReportLab, NumPy, Pandas, Matplotlib
Key Features Implemented	User Authentication, Audio Upload & Recording, Speech-to-Text Transcription, Semantic Similarity Analysis, Audio Feature Extraction, Concept Understanding Score, Dashboard Visualization, PDF Report Generation
Pending / Incomplete Features	None

Setup / Run Instructions	Clone the repository → Create a virtual environment → Install dependencies using pip install -r requirements.txt → Run the application using streamlit run app.py
Additional Notes	Uses OpenAI Whisper for transcription, Sentence-BERT for semantic similarity, Librosa for speech analysis, SQLite for data storage, and ReportLab for PDF report generation.

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions / modules	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes